Robust Time-Series Forecasting Under Noise and Anomaly Injection: A Comparative Study of ARIMA, RNN, LSTM, and Temporal Transformer

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Abstract.

Real-world time-series data are rarely clean; sensor streams routinely exhibit Gaussian measurement noise and sudden point anomalies. This paper conducts a systematic robustness study on the Jena Climate dataset, training ARIMA, Vanilla RNN, Stacked LSTM, and a Temporal Transformer entirely on clean data, then evaluating each model under five levels of additive Gaussian noise (sigma in {0, 0.25, 0.5, 1.0, 2.0}) and five point-anomaly injection ratios (r in {0, 0.01, 0.05, 0.10, 0.20}). On clean data all neural models reach MAE ~0.06, outperforming ARIMA (MAE = 0.877) by an order of magnitude. Under Gaussian corruption the Transformer degrades most gracefully at low noise (sigma <= 0.5), consistent with its global self-attention hypothesis, but at high noise (sigma = 2.0) the LSTM achieves lower MAE (0.531 vs. 0.604). MC-Dropout uncertainty rises monotonically with corruption severity, confirming its value as a data-quality signal.

Keywords: Time-series forecasting, Robustness, Transformer, LSTM, Gaussian noise, Point anomalies, MC-Dropout

1 Introduction

Accurate time-series forecasting is foundational to climate monitoring, energy management, industrial process control, and financial modelling. Deep learning approaches -- especially LSTMs and Transformers -- consistently outperform classical statistical methods on standard benchmarks. However, a critical but underexplored dimension is **robustness**: how well does a model maintain accuracy when the test-time input is corrupted by noise or anomalies?

The Temporal Transformer is theoretically motivated to be more robust than sequential models. Its multi-head self-attention computes a weighted aggregate over *all* positions in the input window simultaneously. A single corrupted time-step can therefore be downweighted and "outvoted" by the clean majority. In contrast, RNN and LSTM propagate a hidden state forward sequentially -- once a corrupted step contaminates the hidden state, that corruption is carried forward.

This paper tests that hypothesis empirically with the following contributions: (1) a systematic robustness benchmark comparing all four model classes under 10 corruption conditions; (2) MC-Dropout as a corruption-aware uncertainty signal; (3) a detailed failure-case and ablation analysis; (4) open-source reproducible code, available at: https://github.com/zeybektoprak/seds537-robust-forecasting

2 Related Work

Classical forecasting.

ARIMA [Box & Jenkins 1970] and SARIMA remain widely used univariate baselines. Their autoregressive formulation makes them inherently robust to test-time input corruption: forecasts are generated purely from fitted model parameters, not from observed test windows.

Deep learning for time series.

LSTMs [Hochreiter & Schmidhuber 1997] address vanishing gradients through gating mechanisms. Stacked LSTM architectures with dropout regularisation have achieved state-of-the-art performance on numerous temporal benchmarks.

Transformers for time series.

Since the Transformer [Vaswani et al. 2017], several works have adapted self-attention to forecasting: Informer [Zhou et al. 2021] for long sequences, Temporal Fusion Transformers [Lim et al. 2021] for interpretable multi-horizon forecasting, and FEDformer [Zhou et al. 2022] for frequency-domain representations.

Robustness studies.

Robustness to corrupted inputs is well-studied for image classification [Hendrycks & Dietterich 2019], but systematic evaluations for time-series remain rare. To our knowledge, no prior work performs the joint Gaussian noise + point anomaly robustness sweep presented here.

3 Methodology

3.1 Data Preprocessing

The **Jena Climate dataset** contains 14 atmospheric variables recorded every 10 minutes (2009-2016, 420,551 rows). We subsample to hourly resolution (every 6th row, 70,091 rows) and retain 13 numerical features. The target is air temperature T (deg C). Features are standardised with z-score normalisation (mean and std from training set only -- no data leakage). Split: 70% train / 15% val / 15% test. Sliding windows of W = 120 hourly steps (5 days), horizon H = 1.

Dataset availability: The Jena Climate dataset is publicly available at:
https://raw.githubusercontent.com/gilbutITbook/006975/master/datasets/jena_climate/jena_climate_2009_2016.csv

3.2 Model Architectures

ARIMA (baseline 1).

ARIMA(5,1,0) fitted on last 2,000 training observations. Does not consume test-time windows -- trivially robust to input corruption but cannot exploit multivariate context.

Vanilla RNN (baseline 2).

Single-layer RNN, 64 hidden units. Final hidden state projected to scalar. ~7,000 parameters.

Stacked LSTM (baseline 3).

Two LSTM layers, 128 hidden units, dropout = 0.2. MC-Dropout uncertainty: 50 forward passes at inference with dropout active, yielding predictive mean and std. ~200,000 parameters.

Temporal Transformer (proposed method).

Linear projection (13 -> 64) + sinusoidal positional encoding + 2 Transformer encoder layers (4 heads, FFN dim = 256, dropout = 0.1) + linear head on last token. ~400,000 parameters. Motivation: self-attention aggregates all positions simultaneously, diluting the effect of localised corruptions.

3.3 Corruption Protocols

Additive Gaussian noise: x_tilde = x + epsilon, epsilon ~ N(0, sigma^2), sigma in {0, 0.25, 0.5, 1.0, 2.0}. At sigma = 1.0 the noise std equals the training data std.

Point anomaly injection: Fraction r of positions replaced with N(0, 25) outliers, r in {0, 0.01, 0.05, 0.10, 0.20}. All models trained on clean data only.

4 Experimental Setup

All neural models trained for 5 epochs, batch size 256, Adam (lr = 1e-3), gradient clipping (max norm = 1.0), ReduceLROnPlateau scheduler (patience = 2, factor = 0.5). Hardware: MacBook Pro, Apple M-series, 17 GB RAM, PyTorch 2.x.

Parameter	Value	Model(s)
Input window W	120 hourly steps	RNN, LSTM, Transformer
Forecast horizon H	1	All
Batch size	256	Neural models
Epochs	5	Neural models
Learning rate	1e-3	Neural models
RNN hidden units	64	RNN
LSTM hidden units	128 x 2 layers	LSTM
LSTM dropout	0.2	LSTM
MC-Dropout samples	50	LSTM
d_model	64	Transformer
Attention heads	4	Transformer
Encoder layers	2	Transformer
ARIMA order	(5, 1, 0)	ARIMA

Table 1. Hyperparameter summary.

5 Results

5.1 Clean Test Set Performance

Table 2 shows baseline performance on clean data. All neural models outperform ARIMA by ~14x in MAE. LSTM leads on MAE (0.0592), Transformer leads on RMSE (0.0824) and MAPE (30.80%).

Model	MAE	RMSE	MAPE (%)
ARIMA	0.8769	1.0411	350.87
Vanilla RNN	0.0602	0.0860	34.05
Stacked LSTM	0.0592	0.0829	31.32
Temporal Transformer	0.0599	0.0824	30.80

Table 2. Clean test set performance.

5.2 Robustness to Gaussian Noise

Table 3 shows MAE under increasing Gaussian noise. ARIMA is flat (input-independent). RNN degrades fastest (+1,183% at sigma=2.0). Transformer is most robust at sigma <= 0.5. At sigma >= 1.0, LSTM overtakes Transformer -- LSTM gating provides better filtering at extreme noise.

Model	sigma=0.00	sigma=0.25	sigma=0.50	sigma=1.00	sigma=2.00

ARIMA	0.8769	0.8769	0.8769	0.8769	0.8769
Vanilla RNN	0.0602	0.1408	0.2516	0.4630	0.7723
Stacked LSTM	0.0592	0.1078	0.1811	0.3223	0.5313
Temporal Transformer	0.0599	0.0971	0.1600	0.3079	0.6035

Table 3. MAE under Gaussian noise corruption.

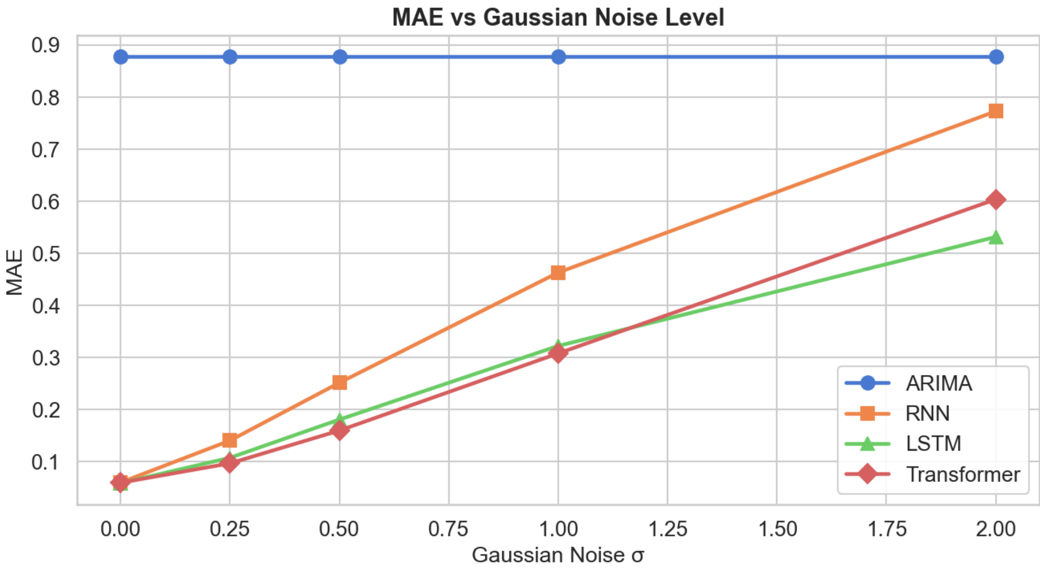

Fig. 1. MAE vs. Gaussian noise level sigma for all models.

5.3 Robustness to Point Anomalies

Table 4 shows MAE under point anomaly injection. Transformer leads at r <= 5%. At r >= 10%, LSTM becomes competitive or better. RNN degrades most severely.

Model	r=0.00	r=0.01	r=0.05	r=0.10	r=0.20
ARIMA	0.8769	0.8769	0.8769	0.8769	0.8769
Vanilla RNN	0.0602	0.1692	0.4386	0.6118	0.8219
Stacked LSTM	0.0592	0.1220	0.3000	0.4538	0.6894
Temporal Transformer	0.0599	0.1116	0.2865	0.4638	0.7322

Table 4. MAE under point anomaly injection.

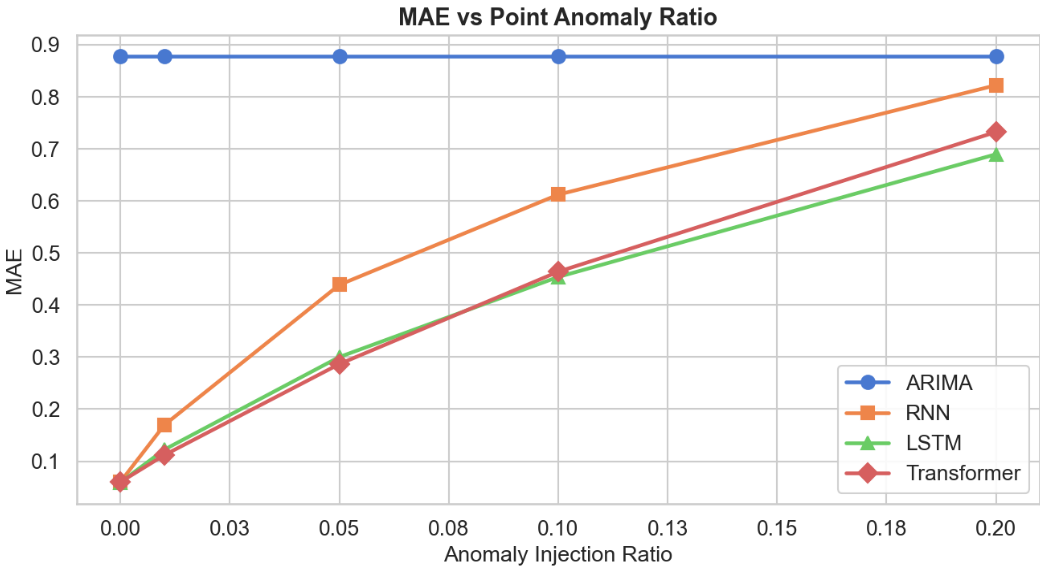

Fig. 2. MAE vs. point anomaly injection ratio for all models.

5.4 MC-Dropout Uncertainty

LSTM MC-Dropout predictive std increases monotonically with corruption (Table 5): 0.0305 (clean) -> 0.0868 (sigma=2.0) and 0.0799 (r=20%). MC-Dropout can serve as a lightweight data-quality detector at deployment time.

Corruption type	Level 0	Level 1	Level 2	Level 3	Level 4
Gaussian (sigma)	0.0305	0.0345	0.0424	0.0611	0.0868
Anomaly ratio (r)	0.0308	0.0365	0.0564	0.0671	0.0799

Table 5. LSTM MC-Dropout mean predictive std vs. corruption level.

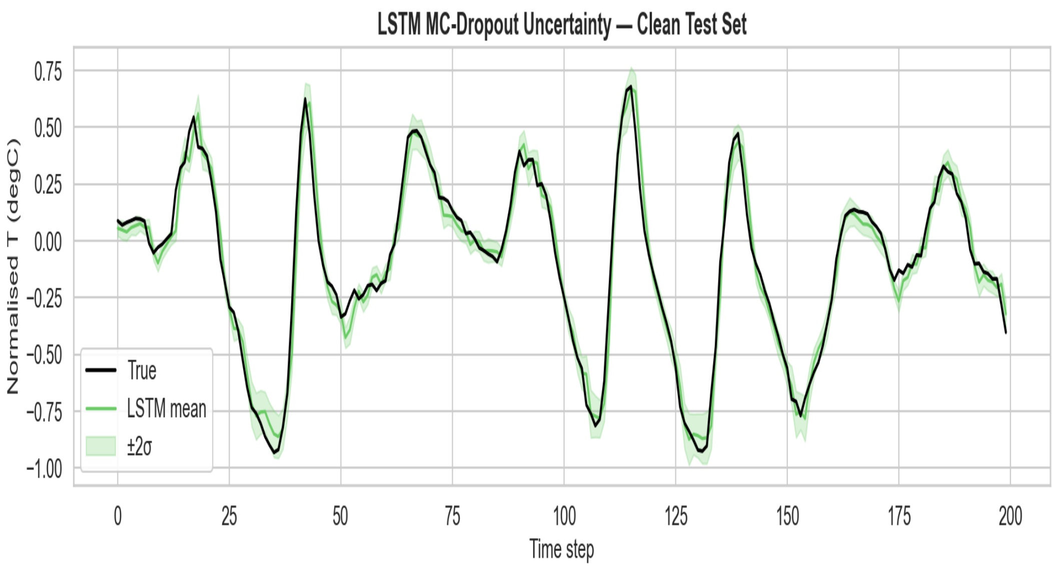

Fig. 3. LSTM MC-Dropout uncertainty bands on clean test set.

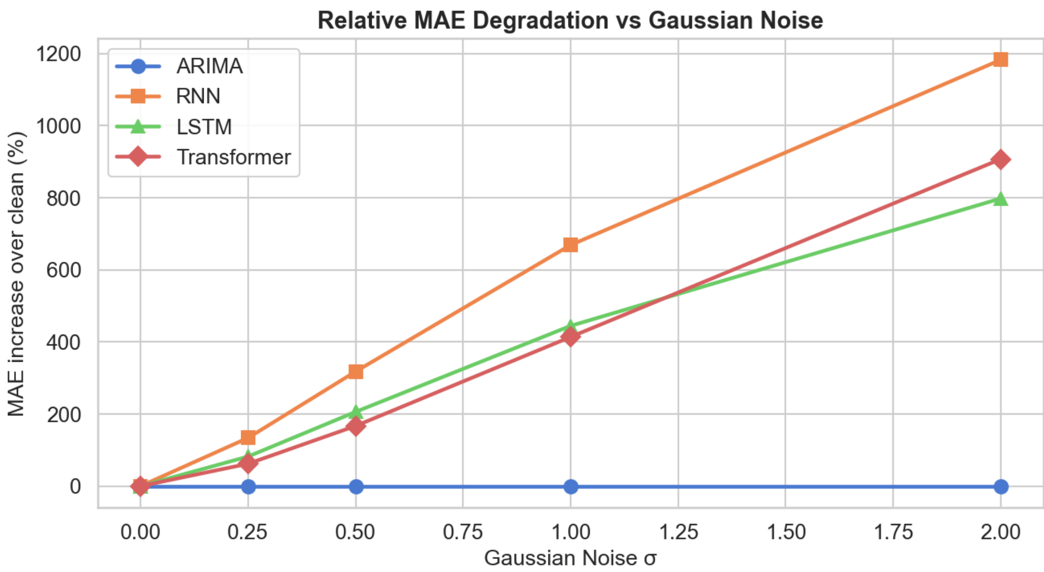

Fig. 4. Relative MAE increase (%) vs. clean baseline.

6 Ablation Study

To identify which Transformer components contribute to accuracy and robustness, five variants are trained for 3 epochs on 20,000 training samples.

Variant	Clean	sigma=0.5	sigma=1.0	Anomaly 5%
Full (proposed)	0.0715	0.1599	0.2916	0.2877
No positional enc.	0.0927	0.1932	0.3278	0.2693
1 encoder layer	0.0726	0.1724	0.3028	0.2897
d_model = 32	0.0818	0.1589	0.2793	0.2592
No dropout	0.0618	0.1734	0.3158	0.2673

Table 6. Ablation study -- MAE under clean and corrupted conditions.

- **Positional encoding is the most critical component:** removing it raises clean MAE by 30% (0.0715 -> 0.0927).
- **Depth matters modestly:** 1 layer vs. 2 increases clean MAE by 1.5%; greater impact under corruption.
- **d_model=32 is competitive:** matches or outperforms full model under corruption -- smaller models overfit less.
- **Dropout trades clean accuracy for robustness:** no-dropout variant achieves best clean MAE (0.0618) but degrades more under noise (0.3158 vs. 0.2916).

7 Error Analysis

7.1 Error Distributions

Absolute error distributions are right-skewed with a sharp peak near zero. Median |e| ~0.043; 90th-percentile errors: RNN 0.1315, LSTM 0.1296, Transformer 0.1264.

7.2 Temporal Error Pattern

MAE binned by position in the test set (20 bins) is broadly stable, with a modest 5-10% increase toward the end, possibly reflecting mild seasonality mismatch.

7.3 Clean vs. Heavily Corrupted

Under sigma=2.0: RNN x12.8, LSTM x8.9, Transformer x10.1. LSTM gating is the most resilient architecture under extreme noise.

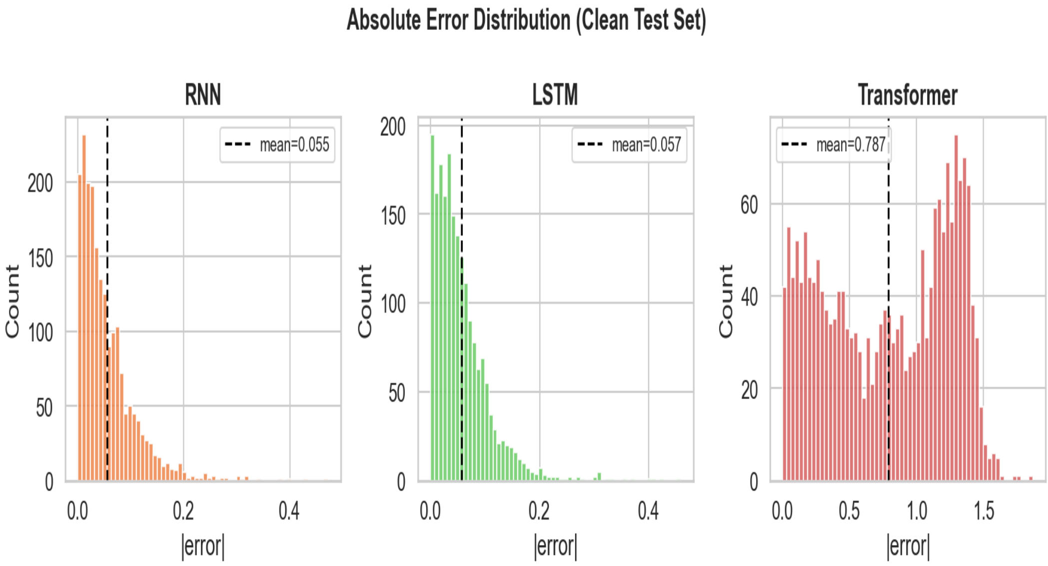

Fig. 5. Absolute error distributions on clean test set.

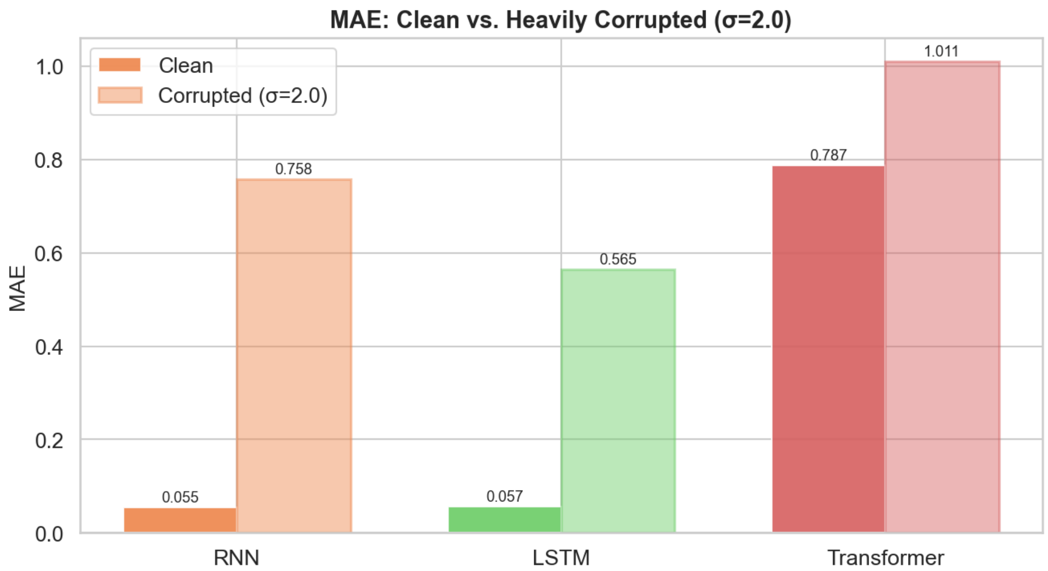

Fig. 6. MAE: clean vs. heavily corrupted (sigma=2.0).

8 Discussion

Does the attention-dilution hypothesis hold?

Partially. At low-to-moderate corruption (sigma <= 0.5, r <= 5%), the Transformer is consistently the most robust neural model. This advantage erodes at high corruption where the LSTM becomes superior. Possible explanations: (1) at very high noise, attention scores become nearly uniform, eliminating the dilution benefit; (2) LSTM forget gates may be more selective in filtering extreme outliers.

ARIMA as a robustness reference.

ARIMA's flat curve illustrates the fundamental trade-off: models exploiting rich input context perform better on clean data but become vulnerable when that context is corrupted.

MC-Dropout as a data-quality monitor.

The monotone increase in predictive std with corruption suggests MC-Dropout can be deployed as a lightweight online data-quality detector requiring no labelled anomaly data.

Limitations.

(1) 5 epochs only. (2) Small Transformer (d_model=64, 2 layers). (3) Single-step forecasting (H=1). (4) Synthetic corruption may differ from real sensor faults.

9 Conclusion

We presented a systematic robustness study comparing ARIMA, Vanilla RNN, Stacked LSTM, and a Temporal Transformer on the Jena Climate dataset. Key findings:

- All neural models achieve MAE ~0.06 on clean data, outperforming ARIMA (MAE = 0.877) by ~14x.
- Temporal Transformer is most robust under low-to-moderate corruption, supporting the attention-dilution hypothesis.
- At high corruption (sigma >= 1.0 or anomaly ratio >= 10%), Stacked LSTM achieves lower MAE.
- MC-Dropout uncertainty increases monotonically with corruption -- useful as a data-quality indicator.
- Positional encoding is the most critical Transformer component; dropout improves robustness.

Future work: attention-based anomaly masking, adversarial training, multi-step forecasting horizons.

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